

Alex Junior Gomez Saltachin

PhD Student in Mathematics, PUC-Rio, Brazil

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Research Interests

Algebraic geometry; birational geometry of Fano and log del Pezzo surfaces; derived categories and semiorthogonal decompositions; exceptional collections; homological mirror symmetry; Landau-Ginzburg models and periods; arithmetic aspects of mirror symmetry.

Current Research

- Big quantum cohomology for log del Pezzo surfaces, with X_{10} in $P(1,2,3,5)$ as a central example.
- Full exceptional collections for canonical stacks of complex log del Pezzo surfaces with $1/3(1,1)$ -singularities, with X_{10} as a central example.

Preprints and Research

- *Exceptional Collections for Canonical Stacks of Log del Pezzo Surfaces with $1/3(1,1)$ -Singularities*. Preprint, arXiv:2606.18238, 2026. Submitted to *Advances in Geometry* (De Gruyter).
- *A Seven-Parameter Landau-Ginzburg Mirror for the Cubic Surface*. In preparation.
- *Real Bruhat Intersections and Complex Open Richardson Varieties*. Draft communication, 2026.

Education

- PhD in Mathematics, PUC-Rio, Brazil, 2023-present. Research advisor: Sergey Galkin. Co-advisor: Vasily Golyshev.
- MSc in Mathematics, Pontificia Universidad Catolica del Peru, 2019. Thesis: Modular Forms. Advisor: Alfredo Poirier.
- BSc in Mathematics, First Class Honors, Pontificia Universidad Catolica del Peru, 2015. Thesis: Elliptic Curves: Mordell-Weil, Roth and Siegel Theorems.

Honors and Scholarships

- FAPERJ Nota 10 Excellence Scholarship, Rio de Janeiro, Brazil, 2024-present.
- First Class Honors, BSc in Mathematics, Pontificia Universidad Catolica del Peru, 2015.

Selected Schools and Programs

- Rutgers Symplectic Summer School (RSSS 2026), Rutgers University, New Brunswick, USA, 2026 - participant.
- AGRA V: School on Arithmetic, Groups and Analysis, Popayan, Colombia, 2026 - Teaching Assistant.
- GARLAT: Latin American School in Arithmetic Geometry, Santiago, Chile, 2026 - participant.
- RGAS Summer School: Mirror Symmetry and Homological Methods in Algebraic Geometry, Zaragoza, Spain, 2026 - participant and poster presenter.
- ICTP School and Workshop on Explicit Arithmetic Geometry, Trieste, Italy, 2025.
- The Mordell Conjecture 100 Years Later, MIT, USA, 2024.
- ICTP String-Math, Trieste, Italy, 2024.
- CIMPA ELGA - Latin American School of Algebraic Geometry, 2024.
- Arizona Winter School, 2022 and 2023; IHES Summer School on the Langlands Program, 2022.
- ICTP-CIMPA Schools in Arithmetic Geometry, 2015 and 2018.

Selected Talks and Posters

- Brauer-Manin and a log del Pezzo surface, Pontificia Universidad Catolica de Chile, 2026.
- Exceptional Collections for Canonical Stacks of Log del Pezzo Surfaces with $1/3(1,1)$ -Singularities, poster, RGAS Summer School, Zaragoza, 2026.
- Periods in Homological Mirror Symmetry and Kloosterman Motives, ICTP, 2025.
- A Tentative Relationship Between Homological Mirror Symmetry and Arithmetic Geometry, poster, Arithmetic and p-adic Geometry in Chile, 2024.
- Mordell-Weil Theorem, PUC-Rio Friday Seminar, 2024.
- Karoubian Completion of Triangulated Categories, PUC-Rio Friday Seminar, 2024.
- Model structures, PUC-Rio Friday Seminar, 2024.
- Delzant's theorem on toric symplectic manifolds, PUC-Rio Friday Seminar, 2023.

Teaching Experience

Graduate instructor and teaching assistant at PUC-Rio and PUCP, 2013-2026. Courses taught or assisted include Linear Algebra, Algebra, Real Analysis, Complex Analysis, Calculus, and Mathematics for Economists. Teaching Assistant at AGRA V: School on Arithmetic, Groups and Analysis, Popayan, Colombia, 2026.

Academic Service

- Organizer of the Etale Cohomology Seminar 2026, a five-week online reading and discussion seminar following Daniel Litt's course *Etale Cohomology and the Weil Conjectures*, starting August 28, 2026. Seminar site: <https://alequisgs.github.io/etale-cohomology-seminar-2026-/>. Invitation PDF available on my website.
- Typesetter, with Mohamed Aliouane, for Masha Vlasenko's lecture notes *Hypergeometric Functions*, ICTP Number Theory and Physics Proceedings, in progress.
- Teaching assistant for *Metodos explicitos de Chabauty: un punto de vista computacional a los puntos racionales*, AGRA V, Popayan, Colombia, 2026.
- Co-founder and organizer of the 38 Movimientos Seminar.
- Co-organizer of the Meetings on Arithmetic and Algebraic Number Theory Seminar.

Languages

Spanish (native); English (advanced); Portuguese (advanced); basic Chinese and French.

Research Mentors

- Sergey Galkin - research advisor, PUC-Rio.
- Vasily Golyshev - co-advisor.

Updated August 2026.